

Hands on metabarcoding - Taxonomy

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Taxonomy

For the taxonomy analysis, we will use **dplyr** and **ggplot2**. Many phyloseq functions for taxonomy use these packages underneath.

Relative abundance (1)

Let's start by transforming Abundance into Relative Abundance, per sample.

Calculate Relative Abundance, using the **Abundance** column:

```
# Calculate relative abundance, per sample
ASVs_full <- ASVs_full %>%
  group_by(Sample) %>%
  ____ (RelativeAbundance = ____*100/sum(____))
```

Relative abundance (2)

Solution:

```
# Calculate relative abundance, per sample  
ASVs_full <- ASVs_full %>%  
  group_by(Sample) %>%  
  mutate(RelativeAbundance = Abundance*100/sum(Abundance))
```

Plot Kingdom level information (1)

Usually, Kingdom/Domain level is not interesting information, but we will use it as a starting point.

Using ggplot2, make a bar plot with **Sample** on x-axis, **RelativeAbundance** on the y-axis and fill the bars based on the **Kingdom**:

```
ASVs_full %>%  
  ggplot(aes(x = _____, y = _____, fill = _____)) +  
  geom_col() +  
  theme_bw() +  
  theme(axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 14),  
        legend.position = "top") +  
  labs(y = "Relative abundance (%)")
```

Plot Kingdom level information (2)

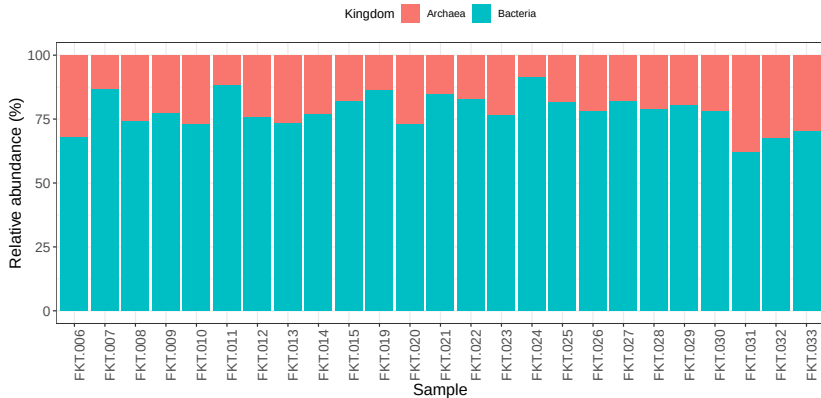
Solution:

```
ASVs_full %>%  
  ggplot(aes(x = Sample, y = RelativeAbundance, fill = Kingdom)) +  
  geom_col() +  
  theme_bw() +  
  theme(axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 14),  
        legend.position = "top") +  
  labs(y = "Relative abundance (%)")
```

See the plot in the next slide.

Plot Kingdom level information (3)

Why the white lines?



Plot Kingdom level information (4)

To solve the white lines, combine each Kingdom relative abundance per sample, using the **summarise** function.

Additionally, add an horizontal line at 75% relative abundance, using **geom_hline()**.

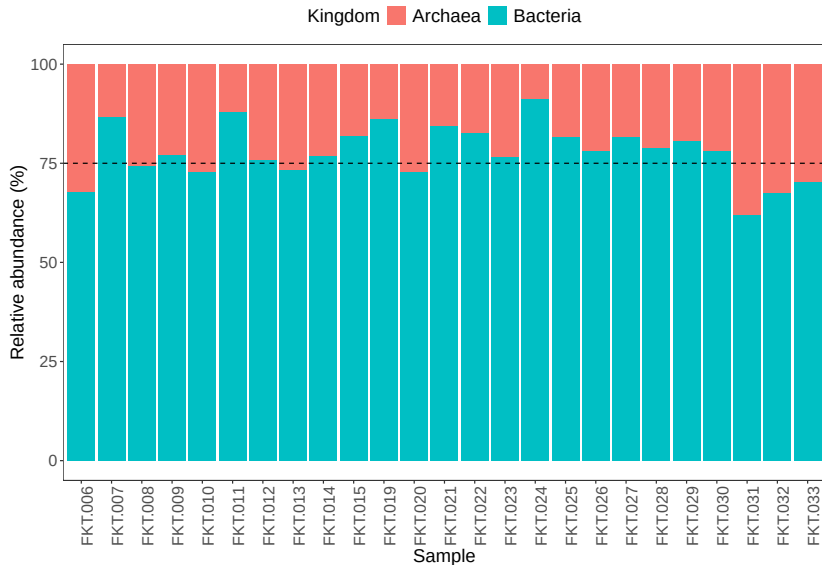
```
ASVs_full %>%
  group_by(Sample, Kingdom) %>%
  ____ (RelativeAbundance = sum(____)) %>%
  ggplot(aes(Sample, RelativeAbundance, fill = Kingdom)) +
  geom_col() +
  geom_hline(yintercept = ____, lty = "dashed") +
  theme_bw() +
  theme(axis.text.x = element_text(size = 14, angle = 90),
        axis.text.y = element_text(size = 14),
        axis.title = element_text(size = 14),
        legend.text = element_text(size = 16),
        legend.title = element_text(size = 16),
        panel.grid = element_blank(),
        legend.position = "top") +
  labs(y = "Relative abundance (%)")
```


Plot Kingdom level information (5)

Solution:

```
ASVs_full %>%  
  group_by(Sample, Kingdom) %>%  
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%  
  ggplot(aes(Sample, RelativeAbundance, fill = Kingdom)) +  
  geom_col() +  
  geom_hline(yintercept = 75, lty = "dashed") +  
  theme_bw() +  
  theme(axis.text.x = element_text(size = 14, angle = 90),  
        axis.text.y = element_text(size = 14),  
        axis.title = element_text(size = 16),  
        legend.text = element_text(size = 16),  
        legend.title = element_text(size = 16),  
        panel.grid = element_blank(),  
        legend.position = "top") +  
  labs(y = "Relative abundance (%)")
```

Plot Kingdom level information (6)



Plot Kingdom level information (7)

The sample ID is not informative, so now we will adapt the previous figure to the experimental design.

Divide the plot in grids, based on **Experiment** and **Replicate**:

```
ASVs_full %>%
  group_by(Sample, Experiment, Treatment, Concentration, Replicate, Kingdom) %>%
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%
  ggplot(aes(x = Treatment, y = RelativeAbundance,
             fill = Kingdom)) +
  geom_col() +
  facet_grid(rows = c(____, ____)) +
  theme_bw() +
  theme(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 14),
        legend.text = element_text(size = 14), legend.title = element_text(size = 14),
        panel.grid = element_blank(),
        legend.position = "top",
        strip.background = element_blank(), strip.text = element_text(size = 14),
        labs(y = "Relative abundance (%)")
```

Plot Kingdom level information (8)

Solution:

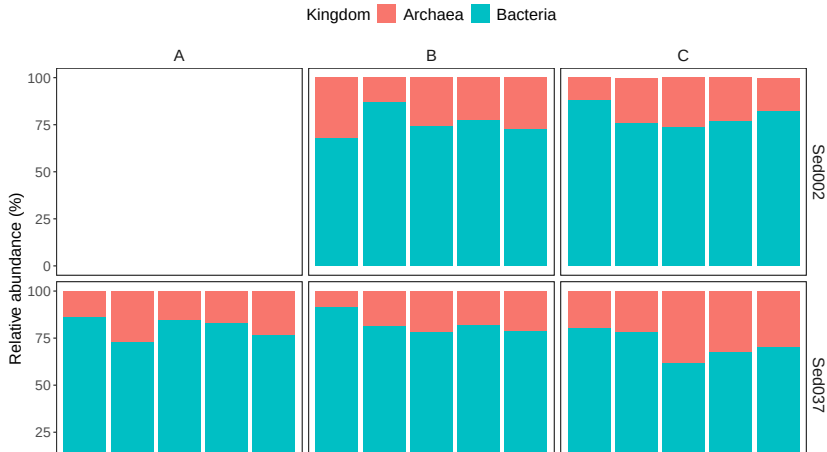
```
ASVs_full %>%
  group_by(Sample, Experiment, Treatment,
            Concentration, Replicate, Kingdom) %>%
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%
  ggplot(aes(x = Treatment, y = RelativeAbundance,
             fill = Kingdom)) +
  geom_col() +
  facet_grid(rows = c("Experiment", "Replicate")) +
  theme_bw() +
  theme(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 14),
        legend.text = element_text(size = 14),
        legend.title = element_text(size = 14),
        panel.grid = element_blank(),
        legend.position = "top",
        strip.background = element_blank(),
        strip.text = element_text(size = 14)) +
  labs(y = "Relative abundance (%)")
```

Plot Kingdom level information (9)

```
## Warning in min(x): no non-missing arguments to min; returning Inf
```

```
## Warning in max(x): no non-missing arguments to max; returning -Inf
```

```
## Warning in min(d[d > tolerance]): no non-missing arguments to min; returning  
## Inf
```



Plot Kingdom level information (10)

Replace the bar plot with a line plot, using grids for the Experiment, but not for the Replicate.

Recall that you can use **geom_point()** for points and **geom_line()** for lines.

```
ASVs_full %>%
```

```
  group_by(Sample, Experiment, Treatment, Concentration, Replicate, Kingdom) %>%
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%
  ggplot(aes(x = Treatment, y = RelativeAbundance, col = Kingdom)) +
  # add a layer with points
  ____ (size = 2.5) +
  # add a layer with lines
  ____ (aes(group = paste(Replicate, Kingdom)))+
  geom_vline(xintercept = 1.5, lty = "dashed") +
  facet_grid(~____) +
  theme_bw() +
  theme(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 14),
        legend.position = "top",
        legend.text = element_text(size = 14),
        legend.title = element_text(size = 14),
        strip.background = element_blank(),
        strip.text = element_text(size = 14),
```

Plot Kingdom level information (11)

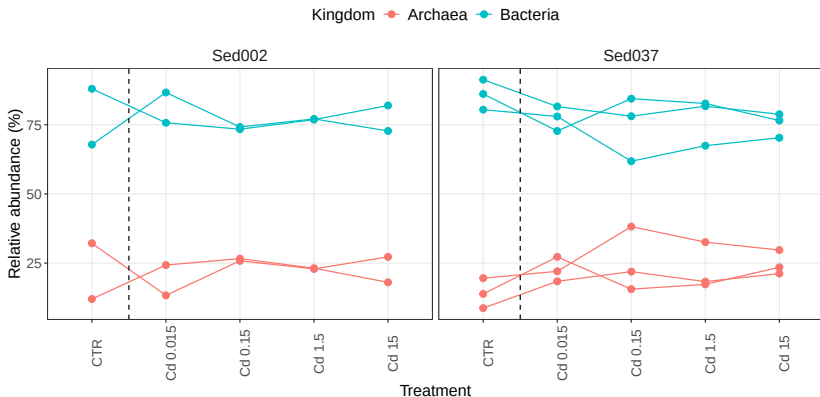
Solution:

```
ASVs_full %>%
```

```
  group_by(Sample, Experiment, Treatment, Concentration, Replicate, Kingdom) %>%
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%
  ggplot(aes(Treatment, RelativeAbundance, col = Kingdom)) +
  geom_point(size = 2.5) +
  geom_line(aes(group = paste(Replicate, Kingdom))) +
  geom_vline(xintercept = 1.5, lty = "dashed") +
  facet_grid(~Experiment) +
  theme_bw() +
  theme(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 14),
        legend.position = "top",
        legend.text = element_text(size = 14),
        legend.title = element_text(size = 14),
        strip.background = element_blank(),
        strip.text = element_text(size = 14),
        panel.grid.minor = element_blank()) +
  labs(y = "Relative abundance (%)")
```

Plot Kingdom level information (12)

This new plot is more informative, because it already establishes that Sediment 2 and 37 had a different response at Kingdom level. In fact, **Bacteria decreased** and **Archaea increased** after Cd treatment.



Plot Phylum level (1)

From the Phylum level and below (Class, etc), there are too many categories to make a good stacked bar plot.

Let's plot only the top 5 phyla, to ensure a satisfactory visualization.

Find the 5 phyla with most relative abundance over the **entire dataset**:

```
top5_phyla <- ASVs_full %>%  
  group_by(____) %>%  
  summarise(totalAbundance = sum(Abundance)) %>%  
  arrange(desc(____)) %>%  
  head(____) %>%  
  pull(Phylum)
```

Plot Phylum level (2)

Solution:

```
top5_phyla <- ASVs_full %>%  
  group_by(Phylum) %>%  
  summarise(totalAbundance = sum(Abundance)) %>%  
  arrange(desc(totalAbundance)) %>%  
  head(n = 5) %>%  
  pull(Phylum)  
top5_phyla
```

```
## [1] "Proteobacteria" "Crenarchaeota" "Planctomycetota" "Bacteroidota"  
## [5] "Chloroflexi"
```

Plot Phylum level (3)

Prepare the data for the stacked bar plot of phylum.

Using **dplyr**, make a new column with the top 5 phyla, and attribute the name **Other** for all phyla that did not make it to the top 5 phyla.

```
top_phylum_data <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate, Phylum) %>%  
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%  
  ____ (topPhyla = ifelse(Phylum %in% top5_phyla, Phylum, ____)) %>%  
  ____ (topPhyla = factor(____, levels = c(top5_phyla, "Other")))
```

Plot Phylum level (4)

Solution:

```
top_phylum_data <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate, Phylum) %>%  
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%  
  mutate(topPhyla = ifelse(Phylum %in% top5_phyla, Phylum, "Other")) %>%  
  mutate(topPhyla = factor(topPhyla, levels = c(top5_phyla, "Other")))
```

```
## `summarise()` has grouped output by 'Sample', 'Experiment', 'Treatment',  
## 'Concentration', 'Replicate'. You can override using the `.groups` argument.
```

Plot Phylum level (5)

Make the plot, using the **top_phylum_data**, and save it as **stacked_bar_plot_phylum**:

```
----- <- ----- +
  ggplot(aes(x = Treatment, y = RelativeAbundance, fill = topPhyla)) %>%
  geom_col() +
  facet_grid(rows = c("Experiment", "Replicate"))
```

Plot Phylum level (6)

Solution:

```
stacked_bar_plot_phylum <- top_phylum_data %>%  
  ggplot(aes(x = Treatment, y = RelativeAbundance, fill = topPhyla)) +  
  geom_col() +  
  facet_grid(rows = c("Experiment", "Replicate"))
```

Plot Phylum level (7)

Add editing layers to the **stacked_bar_plot_phylum** object:

```
----- +
  theme_bw() +
  ----(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 12),
        legend.position = "top",
        strip.background = element_blank(),
        strip.text = element_text(size= 12)) +
  ----(y = "Relative abundance (%)",
        fill = "Top phyla") +
  scale_fill_manual(values = c(qualitative_colors[1:5], "grey80"))
```

Plot Phylum level (8)

Solution:

```
stacked_bar_plot_phylum +  
  theme_bw() +  
  theme(axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 12),  
        legend.position = "top",  
        strip.background = element_blank(),  
        strip.text = element_text(size = 12)) +  
  labs(y = "Relative abundance (%)",  
       fill = "Top phyla") +  
  scale_fill_manual(values = c(qualitative_colors[1:5], "grey80"))
```


Plot Phylum level (9)

```
## Warning in min(x): no non-missing arguments to min; returning Inf
```

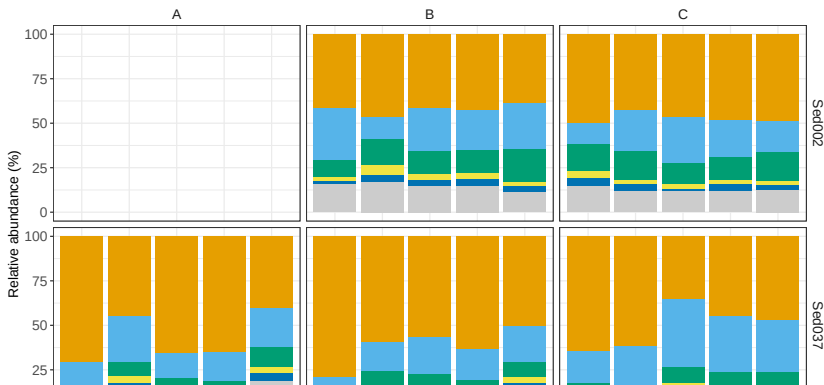
```
## Warning in max(x): no non-missing arguments to max; returning -Inf
```

```
## Warning in min(d[d > tolerance]): no non-missing arguments to min; returning
```

```
## Inf
```

Top phyla

Proteobacteria	Planctomycetota	Chloroflexi
Crenarchaeota	Bacteroidota	Other



Plot Phylum level (10)

Previously, we saw that Bacteria decreased in the Sediment 37. Let's repeat the line plot approach and see if we can identify which phyla were responsible for that.

Use the previous data, **top_phylum_data**, to make a point and line plot:

```
----- %>%
  ggplot(aes(x = Treatment, y = RelativeAbundance, col = topPhyla)) +
  geom_point() +
  geom_line(aes(group = paste(Replicate, topPhyla))) +
  geom_vline(xintercept = 1.5, lty = "dashed") +
  facet_grid(~Experiment) +
  theme_bw() +
  theme(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 14),
        legend.position = "top", legend.text = element_text(size = 12),
        legend.title = element_text(size = 12),
        strip.background = element_blank(),
        strip.text = element_text(size = 14)) +
  labs(y = "Relative abundance (%)", color = "Phyla: ") +
  scale_color_manual(values = c(qualitative_colors[1:5], "grey80"))
```

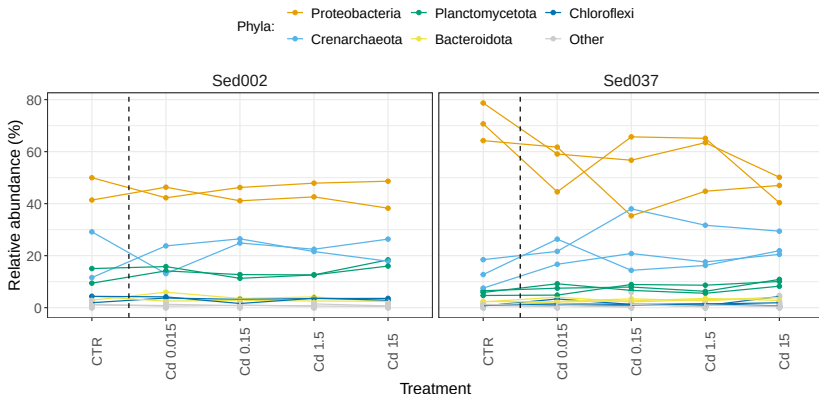
Plot Phylum level (11)

Solution:

```
top_phylum_data %>%
  ggplot(aes(x = Treatment, y = RelativeAbundance, col = topPhyla)) +
  geom_point() +
  geom_vline(xintercept = 1.5, lty = "dashed") +
  geom_line(aes(group = paste(Replicate, topPhyla))) +
  facet_grid(~Experiment) +
  theme_bw() +
  theme(axis.text.x = element_text(size = 12, angle = 90),
        axis.text.y = element_text(size = 12),
        axis.title = element_text(size = 14),
        legend.position = "top",
        legend.text = element_text(size = 12),
        legend.title = element_text(size = 12),
        strip.background = element_blank(),
        strip.text = element_text(size = 14)) +
  labs(y = "Relative abundance (%)",
       color = "Phyla: ") +
  scale_color_manual(values = c(qualitative_colors[1:5], "grey80"))
```

Plot Phylum level (12)

It seems that **Proteobacteria** was the bacterial phylum responsible for the **Bacteria decrease**, while **Crenarchaeota** was the archaeal phylum responsible for the **Archaea increase**.



Plot Genus level (1)

We can obtain more detail within each taxonomic level. The same steps can be used to have a look at the genus level.

Start by finding the 5 most abundant **Genus** in the entire dataset:

```
top5_genera <- ASVs_full %>%  
  group_by(____) %>%  
  summarise(totalAbundance = sum(____)) %>%  
  arrange(desc(____)) %>%  
  head(n = 6) %>%  
  pull(____)
```

Plot Genus level (2)

Solution:

```
top5_genera <- ASVs_full %>%  
  group_by(Genus) %>%  
  summarise(totalAbundance = sum(Abundance)) %>%  
  arrange(desc(totalAbundance)) %>%  
  head(n = 6) %>%  
  pull(Genus)  
top5_genera
```

```
## [1] NA "Stenotrophomonas"  
## [3] "Candidatus Nitrosopumilus" "Woeseia"  
## [5] "Urania-1B-19 marine sediment group" "Colwellia"
```

Plot Genus level (3)

Prepare the data for the plot. To do so, make a new column with the top 5 genera, including an additional category for **Other** and for **Unknown**.

```
top_genus_data <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment, Concentration, Replicate, Genus) %>%  
  summarise(RelativeAbundance = sum(____)) %>%  
  mutate(topGenus = ifelse(____ %in% top5_genera, ____, "Other"),  
         topGenus = ifelse(is.na(topGenus), "Unknown", topGenus)) %>%  
  mutate(topGenus = factor(topGenus, levels = c(____, "Unknown", "Other")))
```

Plot Genus level (4)

Solution:

```
top_genus_data <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate, Genus) %>%  
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%  
  mutate(topGenus = ifelse(Genus %in% top5_genera, Genus, "Other"),  
         topGenus = ifelse(is.na(topGenus), "Unknown", topGenus)) %>%  
  mutate(topGenus = factor(topGenus,  
                           levels = c(top5_genera, "Unknown", "Other")))
```

```
## `summarise()` has grouped output by 'Sample', 'Experiment', 'Treatment',  
## 'Concentration', 'Replicate'. You can override using the `.groups` argument.
```


Plot Genus level (5)

Make a point and line plot, like the previous one, using the **top_genus_data**:

```
---- %>%  
  # set up the ggplot  
  ____ (aes(Treatment, RelativeAbundance, col = topGenus)) +  
  # add point layer  
  ---- +  
  # add line layer  
  ____ (aes(group = paste(Replicate, topGenus))) +  
  # add facet grid  
  ____ (~Experiment) +  
  # use theme black and white  
  ---- +  
  # add theme details manually  
  ____ (axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 14),  
        legend.position = "top",  
        legend.text = element_text(size = 12),  
        legend.title = element_text(size = 12),  
        strip.background = element_blank(),  
        strip.text = element_text(size = 14)) +  
  labs(y = "Relative abundance (%)", color = "Genus: ") +  
  scale_color_manual(values = c(qualitative_colors[1:5], "grey41", "grey80"))
```

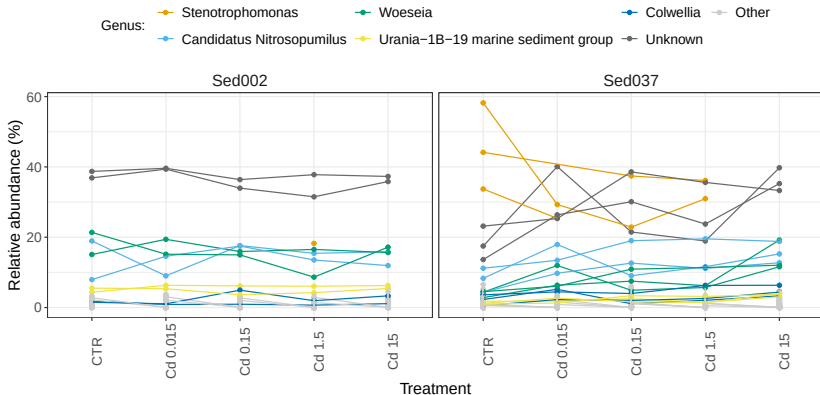
Plot Genus level (6)

Solution:

```
top_genus_data %>%  
  # set up the ggplot  
  ggplot(aes(Treatment, RelativeAbundance, col = topGenus)) +  
  # add point layer  
  geom_point() +  
  # add line layer  
  geom_line(aes(group = paste(Replicate, topGenus))) +  
  # add facet grid  
  facet_grid(~Experiment) +  
  # use theme black and white  
  theme_bw() +  
  # add theme details manually  
  theme(axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 14),  
        legend.position = "top",  
        legend.text = element_text(size = 12),  
        legend.title = element_text(size = 12),  
        strip.background = element_blank(),  
        strip.text = element_text(size = 14)) +  
  labs(y = "Relative abundance (%)",  
       color = "Genus: ") +
```

Plot Genus level (7)

Stenotrophomonas seems to be the responsible for the Bacterial decrease in sediment 37; and the Cd treatment increased the relative abundance of **unknown** genera.



Closer look at *Stenotrophomonas* (1)

Sometimes, it might be better to focus on a single taxonomic group, instead of looking at several at once.

Calculate the relative abundance of each genus, per sample, then filter the ***Stenotrophomonas*** genus and save it as **steno_data** object:

```
steno_data <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate, _____) %>%  
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%  
  filter(_____)
```

Closer look at *Stenotrophomonas* (2)

Solution:

```
steno_data <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate, Genus) %>%  
  summarise(RelativeAbundance = sum(RelativeAbundance)) %>%  
  filter(Genus == "Stenotrophomonas")
```

```
## `summarise()` has grouped output by 'Sample', 'Experiment', 'Treatment',  
## 'Concentration', 'Replicate'. You can override using the `.groups` argument.
```

Closer look at Stenotrophomonas (3)

Using the **steno_data**, plot Stenotrophomonas relative abundance as a function of Cd concentration. Recall that the **Concentration** column includes the Cd concentration as a numerical variable.

Store the plot as **steno_plot**:

```
_____ <- _____ %>%  
ggplot(aes(x = _____,  
           y = _____)) +  
  geom_point() +  
  geom_smooth(se = FALSE,  
             method = "lm",  
             lty = "dashed",  
             col = "black")
```

Closer look at Stenotrophomonas (4)

Solution:

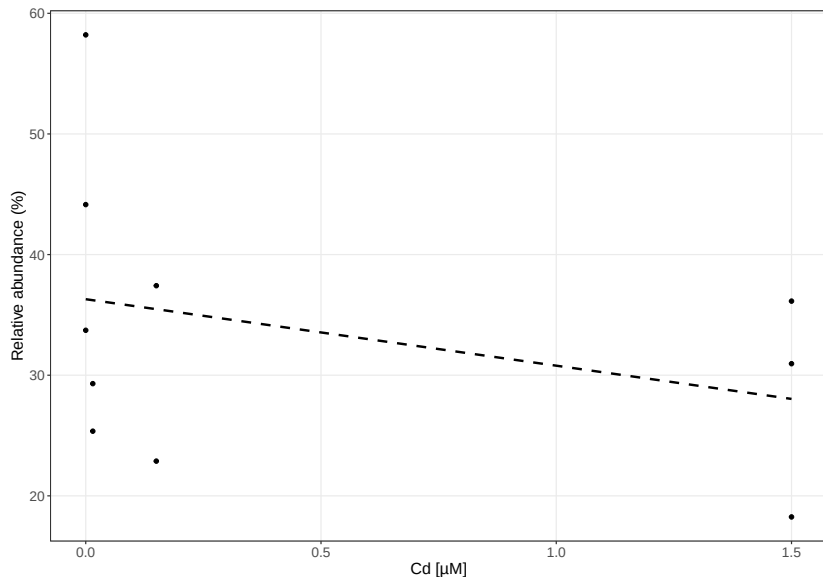
```
steno_plot <- steno_data %>%  
  ggplot(aes(x = Concentration,  
             y = RelativeAbundance)) +  
  geom_point() +  
  geom_smooth(se = FALSE,  
             method = "lm",  
             lty = "dashed",  
             col = "black")
```

Closer look at Stenotrophomonas (5)

Add some editing on top of **steno_plot**:

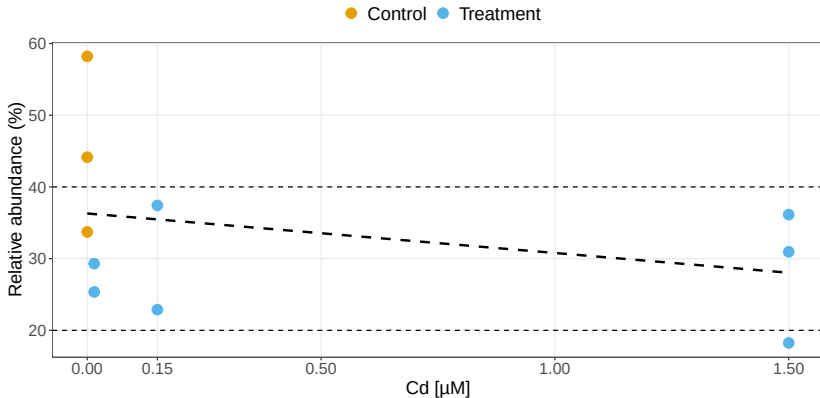
```
steno_plot ----  
  theme_bw() +  
  ----(panel.grid.minor = element_blank(),  
        axis.text = element_text(size = 12),  
        axis.title = element_text(size = 14)) +  
  labs(y = "Relative abundance (%)",  
        x = "Cd [\U00B5M]")
```


Closer look at *Stenotrophomonas* (6)



Closer look at *Stenotrophomonas* (7)

Some improvements, can you do it?



Closer look at Stenotrophomonas (8)

The code:

```
steno_data %>%  
  mutate(TreatmentColor = ifelse(Treatment == "CTR", "Control", "Treatment")) %>%  
  ggplot(aes(x = Concentration,  
            y = RelativeAbundance)) +  
  geom_point(aes(col = TreatmentColor), size = 4) +  
  geom_smooth(se = FALSE,  
            method = "lm",  
            lty = "dashed",  
            col = "black") +  
  theme_bw() +  
  geom_hline(yintercept = c(20,40), lty = "dashed") +  
  theme(panel.grid.minor = element_blank(),  
        axis.text = element_text(size = 14),  
        axis.title = element_text(size = 16),  
        legend.text = element_text(size = 16), legend.title = element_text(size = 16),  
        legend.position = "top") +  
  labs(y = "Relative abundance (%)",  
       x = "Cd [\U00B5M]",  
       color = "") +  
  scale_color_manual(values = qualitative_colors) +  
  scale_x_continuous(limits = c(0,1.5), breaks = c(0, 0.15, 0.5, 1, 1.5))
```